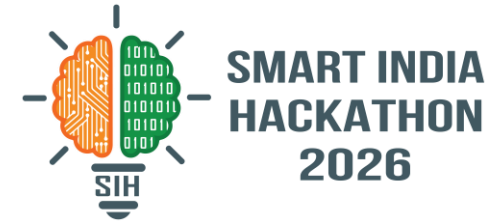


# SMART INDIA HACKATHON 2026



## TITLE PAGE

Problem Statement ID

**SIH2026027**

Problem Statement Title

**Automatic Block Planning for Indian Railways**

Theme • Smart Automation

Category • Software

Team ID • (assigned on SIH portal)

**Team Name • Anomaly**

Galgotias University, Uttar Pradesh



Anomaly

### THE PROBLEM

- Engineering, Signalling and Traction each book track blocks in isolation
- Blocks clash → windows wasted, maintenance deferred, trains delayed

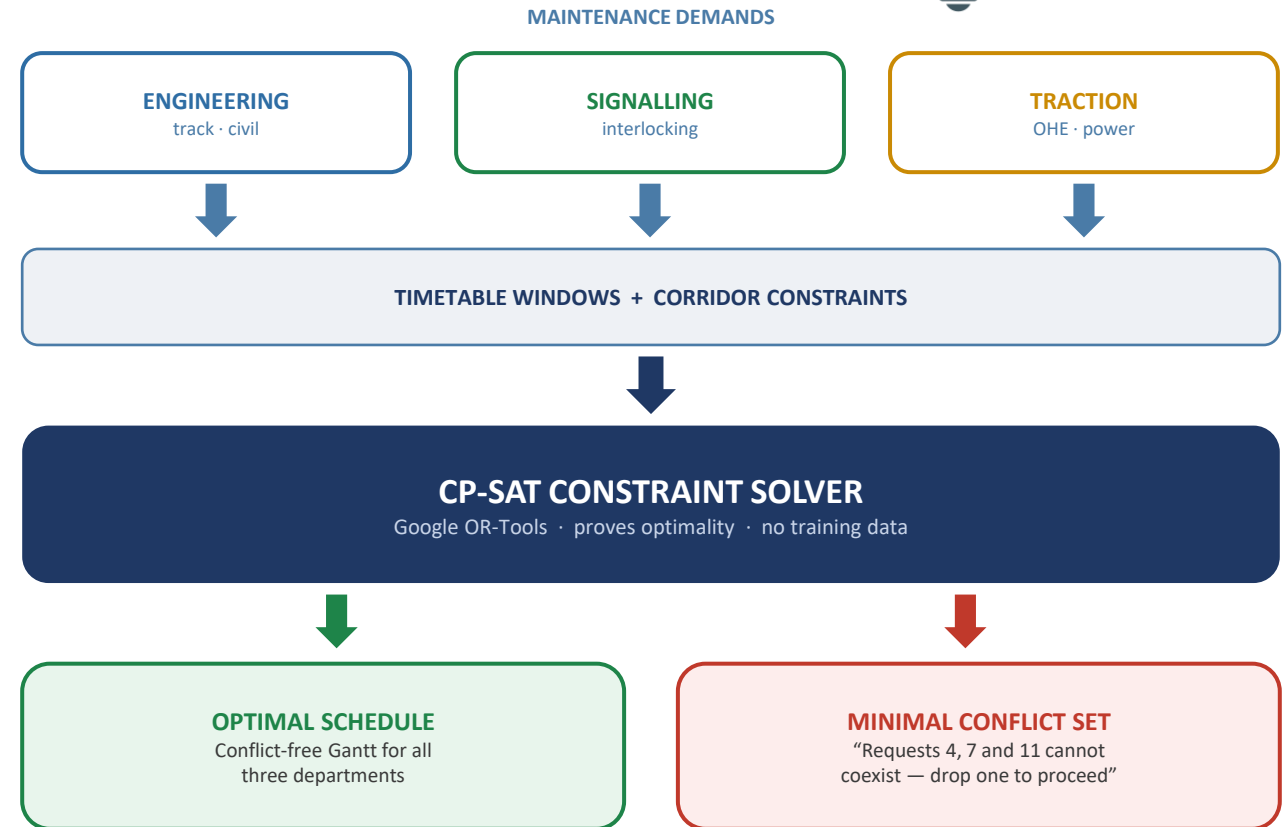
### OUR SOLUTION — RAILBLOCK

- One engine ingests all three departments' demands plus the timetable
- Returns a conflict-free block plan, weekly and monthly

### WHY IT IS DIFFERENT

- **Proven, not predicted — CP-SAT certifies the plan is optimal**
- When demands cannot all fit, it names the exact conflicting requests
- Priority slider re-solves the whole corridor in under 2 seconds
- No training data, no model drift, every decision traceable

# PROPOSED SOLUTION



*Either answer is a proof — never a guess.*

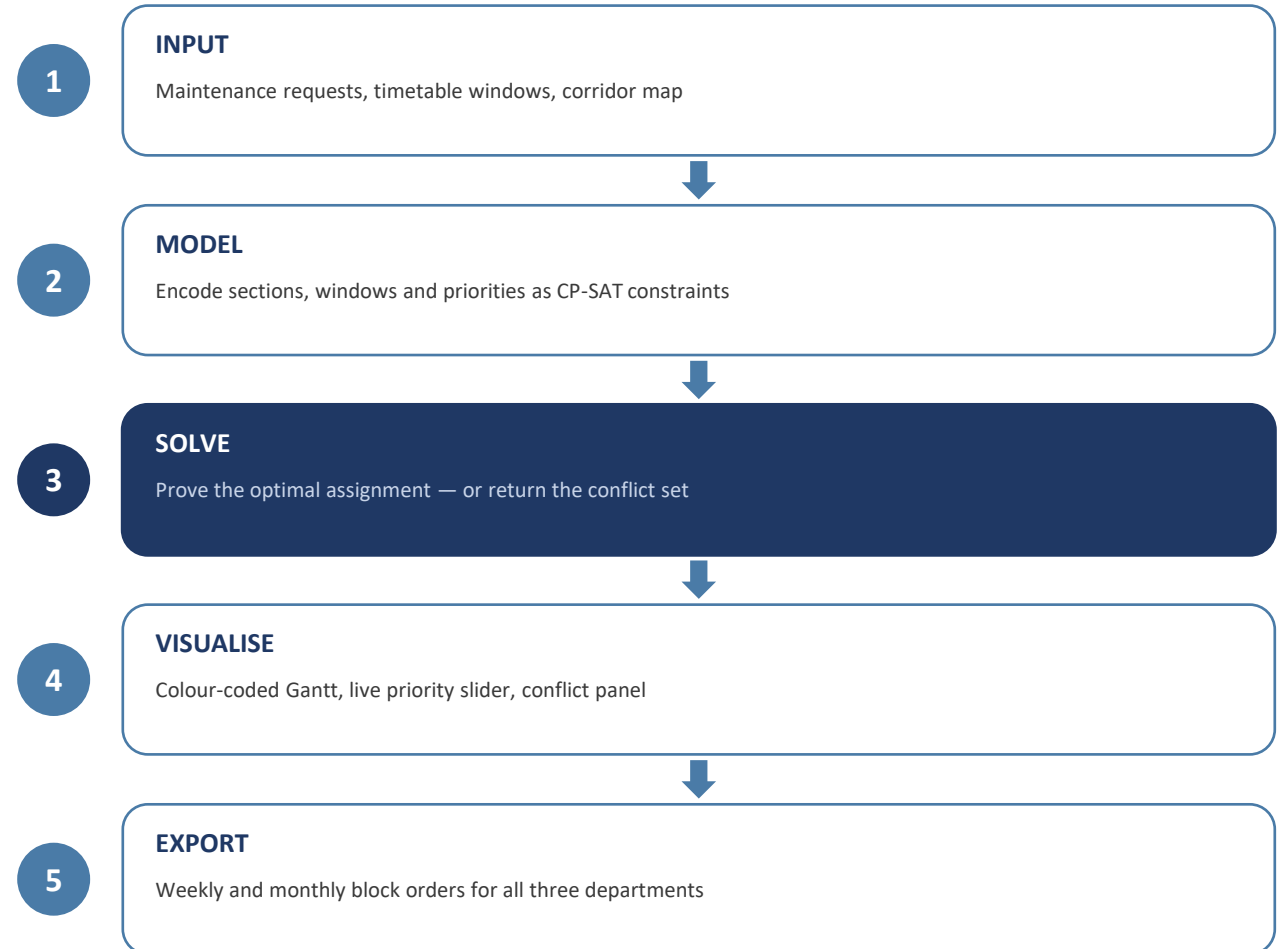
# TECHNICAL APPROACH

## TECHNOLOGY

- Solver — Google OR-Tools CP-SAT (open source)
- Backend — Python 3, FastAPI REST service
- Frontend — React.js with Plotly interactive Gantt
- Data — parameterised corridor scenario generator
- **Runs fully offline on one laptop — no GPU, no cloud**

## CONSTRAINTS MODELLED

- Section exclusivity — one occupant per track section
- Timetable compliance — blocks fit approved windows only
- Duration bounds and inter-block corridor sequencing
- Crew availability per department per day
- Urgency-weighted objective — tunable at run time



# FEASIBILITY AND VIABILITY

## WHY IT IS BUILDABLE

- OR-Tools installs with one pip command — no GPU, no licence fee
- The core solver is roughly 150 lines of Python
- Standard Python + React stack — nothing new to learn mid-build
- **The statement lists its own inputs, so there is no data barrier**

## HOW WE DE-RISK IT

- Week 1 gate: a 50-line CP-SAT model must solve a toy corridor
- Demo one small division and one full corridor to prove scaling
- Partial schedules still render — no all-or-nothing failure
- Six members on parallel tracks: solver, UI, scenarios, domain, pitch

### RISK

No access to live TMS / BDMS / SMMS systems

Railway block-working rules are unfamiliar territory

Only 36 hours to build at the finale

### MITIGATION

Adapter layer documented; parameterised scenarios are accepted method

One member owns the rulebook from week 1 and validates every constraint

Solver proven before the event; six members run independent workstreams

# IMPACT AND BENEFITS



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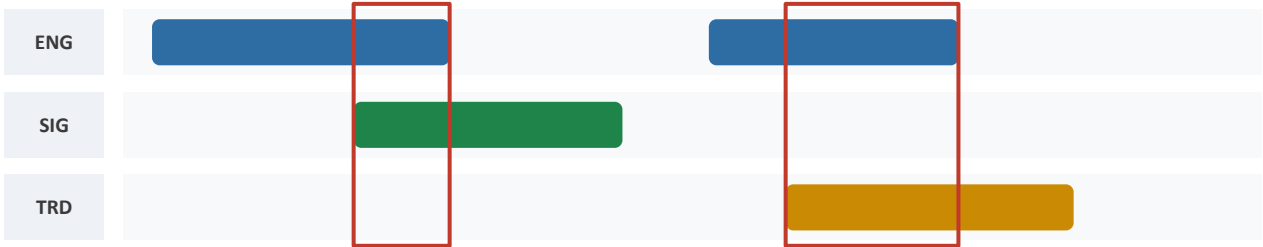
## OPERATIONAL

- Clashes are resolved weeks ahead instead of on the morning itself
- Every approved maintenance window gets filled, none wasted
- Fewer train delays caused by blocks that overrun or collide
- Each decision traces back to a named constraint — fully auditable

## REACH AND SUSTAINABILITY

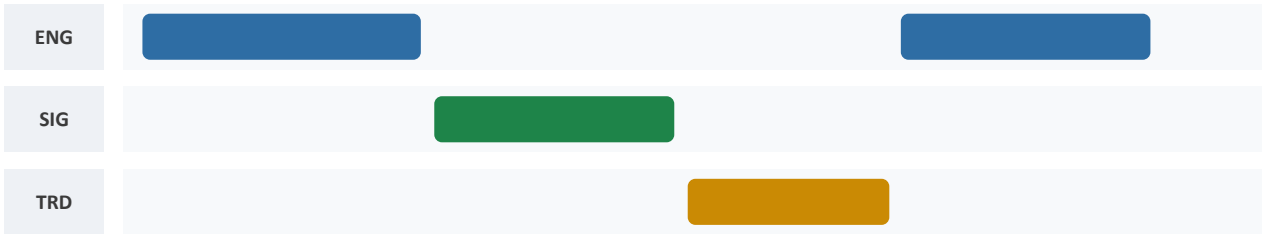
- Scales from one division to all 68 with no change to the code
- Feeds the existing eBlock workflow — no retraining for field staff
- **Zero licensing cost — the entire stack is open source**
- Better-planned maintenance means healthier track and safer running

**TODAY** — each department plans on its own



✖ blocks collide — one gets cancelled on the day

**WITH RAILBLOCK** — one optimised plan



✓ zero conflicts · every window used · proven optimal

**13 M+**

passengers a day

**68**

divisions reachable

**3**

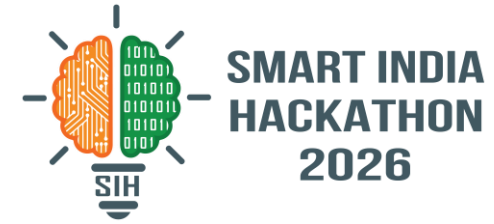
departments unified

**< 2 s**

to re-solve

Anomaly

# RESEARCH AND REFERENCES



## DOMAIN

- Indian Railways — Block Working Rules and departmental circulars
- Ministry of Railways — eBlock system documentation
- NTES — National Train Enquiry System timetable data

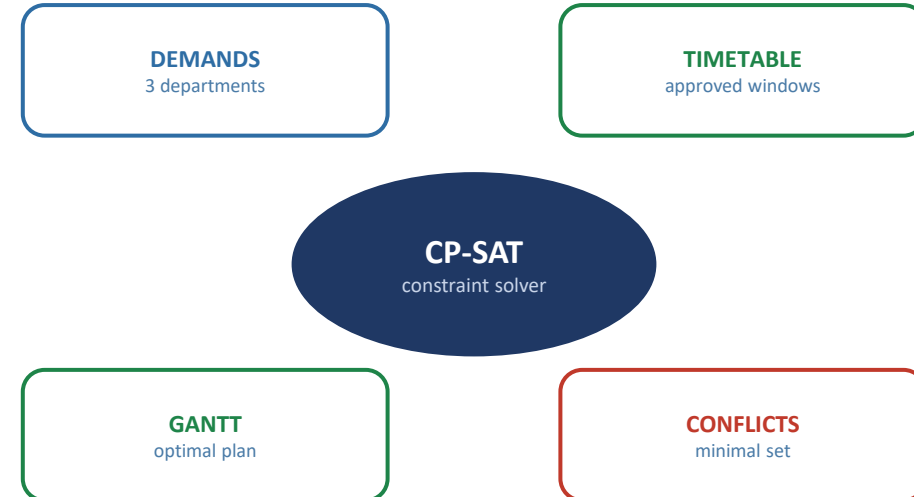
## TECHNICAL

- Google OR-Tools CP-SAT — [developers.google.com/optimization](https://developers.google.com/optimization)
- Caprara, Fischetti & Toth, “Modeling and Solving the Train Timetabling Problem”, Operations Research 50(5), 2002
- Perron & Furnon, OR-Tools, Google LLC, 2023

## PRECEDENT

- **SIH 2025 — AI-Driven Train Induction Planning for KMRL: the same class of railway scheduling problem, solved with optimisation**

### SYSTEM AT A GLANCE



*Four inputs, one solver, two possible answers — a proven schedule or the exact reason one is impossible.*